

This project is a **training site, not a real medical device** under any regulatory definition — it is not Software as a Medical Device (SaMD) or Software in a Medical Device (SiMD). This page is not a genuine IEC 62304 deliverable: it illustrates the *kind* of document a real SaMD/SiMD project would produce. See the [document register](#) for the full explanation.

Clause: 5.3 — Software Architectural Design · **Register:** [Document Register](#)

Clause 5.3 has **no Class A requirement at all** — every row below applies at Class B and C only, one (5.3.5) at Class C alone.

What the standard requires (Class C — every sub-clause of 5.3)

REF	TITLE	APPLIES AT	OUTPUT
5.3.1	Transform requirements into an architecture	B, C	Documented architecture describing the software structure and identifying the software items
5.3.2	Architecture for interfaces of software items	B, C	Documented architecture for interfaces between software items and external components
5.3.3	Specify functional/performance requirements of SOUP	B, C	Functional and performance requirements specified for each SOUP item
5.3.4	Specify system hardware/software required by SOUP	B, C	System hardware and software specified as necessary to support each SOUP item
5.3.5	Identify segregation necessary for risk control	C only	Segregation between software items necessary for risk control, and how its effectiveness is ensured
5.3.6	Verify software architecture	B, C	Documented verification that the architecture implements requirements, supports interfaces, and supports SOUP operation

5.3.1 — Software items

There is no build step and no bundler — every script *is* the software item it defines, one file each:

SOFTWARE ITEM	RESPONSIBILITY
nav.js	Shared navigation — highlights the active page link
async-utils.js	Shared async helpers: <code>delay()</code> , <code>fetchJSON()</code> (centralises the <code>response.ok</code> check <code>fetch</code> doesn't do itself)
theme.js	Light/dark theme toggle and persistence

SOFTWARE ITEM	RESPONSIBILITY
<code>learn.js</code>	Learn page: card rendering, level toggle, safety-class filter, deliverables list, progress tracker
<code>quiz.js</code>	Quiz page: question loading with prefetch, shuffle, timer, scoring, certificate
<code>contact.js</code>	Contact page: field validation, asynchronous submission with timeout
<code>style.css</code>	All visual presentation, including both theme palettes

Each page loads `nav.js`, then `async-utils.js`, then its own page script, all marked `defer` — deferred scripts run in document order, which is what guarantees the shared helpers exist before a page script calls them (the project's design notes record why this ordering was chosen over a bundler).

5.3.2 — Interfaces between software items

There is no shared global state between pages. Where two items do need to communicate, the interface is one of:

INTERFACE	BETWEEN	MECHANISM
Data loading	<code>learn.js</code> / <code>quiz.js</code> ↔ <code>data/*.json</code>	<code>fetch()</code> via <code>fetchJSON()</code> , validated for shape on the way in
Cross-page state	<code>learn.js</code> → <code>quiz.js</code>	<code>localStorage['62304_trainingLevel']</code> — the two pages share no JavaScript and communicate only through this key
Theme state	<code>theme.js</code> ↔ every page	<code>localStorage['62304_theme']</code> plus the <code>data-theme</code> attribute on <code><html></code>
Banner dismissal	<code>learn.js</code> (write) / <code>learn.js</code> (read on reload)	<code>localStorage['62304_bannerDismissed']</code>
Form submission	<code>contact.js</code> → Formspree	<code>fetch()</code> POST to <code>https://formspree.io/f/mpqgydrv</code> , JSON request/response

Every `localStorage` key is namespaced `62304_...` specifically so its owner is unambiguous in DevTools — itself a small interface-design decision worth recording, since an unprefixed key is a common source of collision when a page later gains a second script that also wants "the" storage key.

5.3.3–5.3.4 — SOUP

Nothing third-party ships to a visitor's browser. The production pages load no external script, font, or stylesheet — confirmed by `grep` across all five HTML files, whose only external references are outbound links (to `stjohnlynch.com`, `iec.ch`, `dataprotection.ie`, and the Formspree endpoint itself) rather than resources the page depends on to render. That is a deliberate architectural property, not an oversight, and it is why the SOUP table below is short:

SOUP ITEM	MANUFACTURER	VERSION	SHIPS TO VISITORS?	FUNCTIONAL/PERFORMANCE REQUIREMENT	ENVIRON REQUIRE
Playwright	Microsoft	>=1.40 (see tests/requirements.txt)	No — test-only	Drives a real Chromium/Chrome to exercise the site as a user would	Python 3, Chromiur Chrome i
axe-core	Deque Systems	4.10.2 , pinned by URL in tests/test_site.py (AXE_CDN)	No — test-only	WCAG 2.1 A/AA + best- practice accessibility scanning	Downloa test time; is skippec failed) if unreacha
Formspree is not SOUP in the 5.3 sense — it is not a software component embedded in the product, it is an external HTTP service the contact form calls at runtime, closer to an <i>interfacing system</i> than to third-party code under this project's control. It is documented here for completeness because a reader learning what SOUP means benefits from seeing a real example of something that is deliberately excluded and why. 5.3.5 — Segregation for risk control (Class C only) The one place this project asks "could a mistake here affect something that matters" is the safety-class data. <code>learn.js</code>'s <code>mergeApplicability()</code> segregates <i>rendering</i> from <i>trusting</i>: the merge step is a hard gate — if <code>phases.json</code> and <code>applicability.json</code> disagree, no card renders at all, rather than rendering with unverified data. See 11 — Risk Management File for the incident that made this segregation necessary. 5.3.6 — Architecture verification There is no standalone architecture review record — the closest equivalent is that every interface listed above is exercised directly by <code>tests/test_site.py</code> (for example, the <code>learn</code> and <code>applicability</code> groups assert on the <code>localStorage</code> keys and the render-gate behaviour described above), which verifies the architecture indirectly, by verifying its observable effects, rather than by a separate design review. Gaps • No dedicated architecture diagram exists outside the tables above — for a project this size (seven files) a diagram would likely repeat the file table rather than add information; that trade-off would flip for a larger codebase. • 5.3.6's verification is behavioural, not a formal design review with reviewer sign-off — recorded as what it is rather than dressed up as one.					